

PHY202: Problem Set 6

NOTE: Submit the solution to problem numbers 3(i) and 4(iii) in the beginning of the class on 12th November.

1. Differential equations of the form $y' + P(x)y = Q(x)y^n$ are known as Bernoulli's equations. Show that such non-linear equations can be put in the linear form by the substitution $z = y^{1-n}$.

Use this idea to solve, $3xy' + y + x^2y^4 = 0$.

2. Differential equations of the form $x^2y'' + pxy' + qy = 0$, with p, q as constant coefficients, are known as Euler's equidimensional equations. Find out the result of the substitution $x = e^z$ on such equations.

3. Find a particular solution of each of the following differential equations by variation of parameters (solve the homogeneous equation, assume $y_p = v_1y_1 + v_2y_2$, and find v_1 and v_2):

(i) $y'' + y = \sec x$

(ii) $y'' + y = \sec x \tan x$

4. Find the general solution of each of the following equations (by any method):

(i) $(x^2 - 1)y'' - 2xy' + 2y = (x^2 - 1)^2$

(ii) $(x^2 + x)y'' + (2 - x^2)y' - (2 + x)y = x(x + 1)^2$

(iii) $xy'' - (1 + x)y' + y = x^2e^{2x}$

5. Find a particular solution of each of the following differential equations by using the operator method:

(i) $y'' - 2y' - 3y = 6e^{5x}$

(ii) $y''' - y'' = 12x - 2$

(iii) $y'' + 2y' + y = 2x^2 e^{-2x} + 3e^{2x}$